

**internal-external rotation 90 degree-abducted — in vivo – ex vivo difference, 95% CI (overlap only)**

**Sternoclavicular — Protraction(+)/retraction(-)**

**Sternoclavicular — Depression(+)/elevation(-)**

**Sternoclavicular — Backwards(+)/forward(-) rot**

not modelled

not modelled

not modelled

**Acromioclavicular — Protraction(+)/retraction(-)**

**Acromioclavicular — Medial(+)/lateral(-) rot**

**Acromioclavicular — Posterior(+)/anterior(-) tilt**

not modelled

not modelled

not modelled

**Scapulothoracic — Protraction(+)/retraction(-)**

**Scapulothoracic — Medial(+)/lateral(-) rot**

**Scapulothoracic — Posterior(+)/anterior(-) tilt**

single group

single group

single group

### Glenohumeral — Plane of elevation

**Glenohumeral — Elevation(-)/Depression(+)**

**Glenohumeral — Int(+)/ext(-) rotation**

single group

single group

single group

blue line = in vivo – ex vivo, band = 95% CI · squares along the bottom = band excludes 0 · dashed verticals = interior knots, dots = value at each knot

top-right: one dot per in-vivo difference coefficient, read LEFT TO RIGHT as gamma\_0, gamma\_1, ... — the index is printed under each dot. gamma\_0 = constant level shift (condoin vivo); gamma\_k = ns\_k:condoin vivo, a shape term filled = that coefficient passed |coef/se| >= 1.96, hollow = did not. stars = JOINT test on ALL difference coefficients at once, BH-FDR: \*\*\* p<0.001, \*\* p<0.01, \* p<0.05, ns = not significant

bottom-left: shoulders ex/in · studies ex/in | Wald ref/diff counts | mean and max  $|\Delta|$  in degrees. Curves are drawn only where  $\geq 2$  shoulders have data.